Ch 35 · Acute ischaemic stroke

Reperfusion

- IV alteplase/tenecteplase within 4.5 h; BP < 185/110 before, < 180/105 after
- Thrombectomy for LVO within 6 h; to 24 h with mismatch (DAWN, DEFUSE-3); large core also benefits

ICU care

- No antithrombotics 24 h post-lysis; normoglycaemia, normothermia; swallow screen
- Haemorrhagic transformation: stop lysis, cryoprecipitate/TXA

Malignant MCA infarction

- Oedema peaks day 2–5; hemicraniectomy within 48 h ≤ 60 years ↓ mortality
- DESTINY II (> 60): survival with more disability – discuss values

Key point: Time is brain: door-to-needle and door-to-groin targets.

Ch 36 · Bacterial meningitis

Immediate actions

- Ceftriaxone 2 g 12-hourly ± amoxicillin (> 60/immunocompromised) ± vancomycin
- Dexamethasone 10 mg 6-hourly × 4 days with first dose
- CT before LP only if focal signs, seizures, papilloedema, GCS < 10, immunocompromise

CSF

- Bacterial: neutrophils, protein > 1 g/L, glucose < 40% serum
- Viral: lymphocytes, normal glucose; TB/fungal: lymphocytes, very low glucose

Public health

- Droplet precautions 24 h; notify; ciprofloxacin prophylaxis for close contacts

Key point: Never delay antibiotics for imaging or LP.

Ch 37 · Encephalitis

Causes

- HSV-1, VZV, enterovirus, arboviruses; autoimmune (NMDAR, LGI1); Listeria, TB

Investigation

- CSF HSV PCR (repeat if early negative), autoimmune antibodies, MRI, EEG

Treatment

- IV aciclovir 10 mg/kg 8-hourly immediately, 14–21 days if HSV
- Autoimmune: steroids, IVIG/PLEX, tumour removal, rituximab

ICU complications

- Seizures/NCSE, raised ICP, autonomic instability, SIADH, aciclovir nephrotoxicity

Key point: Start aciclovir on suspicion – do not wait for PCR.

Ch 38 · Intracerebral haemorrhage

Acute care

- Reverse anticoagulation: andexanet (ANNEXA-I) or 4F-PCC; vitamin K for warfarin
- SBP to 140 (not < 110) smoothly; care bundle (INTERACT3)
- Airway if GCS ≤ 8; treat seizures, no prophylaxis

ICH score

- GCS, volume ≥ 30 mL, IVH, infratentorial, age ≥ 80; avoid early DNAR

Surgery

- EVD for hydrocephalus; cerebellar > 3 cm: evacuate
- Minimally invasive evacuation of lobar ICH (ENRICH) may help

Key point: Haematoma expansion is the modifiable target – BP and reversal fast.

Ch 39 · Subarachnoid haemorrhage

Grading

- WFNS (GCS + motor deficit), Hunt–Hess; modified Fisher predicts vasospasm

Early management

- Secure aneurysm early; coiling if feasible (ISAT)
- SBP < 160 before securing; nimodipine 60 mg 4-hourly × 21 days
- EVD for hydrocephalus; euvolaemia, normonatraemia

Delayed cerebral ischaemia

- Day 4–14; new deficit; TCD/CT perfusion
- Euvolaemia, induced hypertension; endovascular rescue; exclude other causes

Key point: No prophylactic triple-H – only induced hypertension for established DCI.

Ch 40 · Status epilepticus

Definitions

- ≥ 5 min seizure or no recovery; established = after benzodiazepines; refractory; super-refractory

Management

- Benzodiazepine (IV lorazepam/IM midazolam) ×2
- Second-line: levetiracetam, valproate or fosphenytoin – equal (ESETT)
- Refractory: intubate, midazolam/propofol/thiopental, continuous EEG
- Find cause: glucose, electrolytes, drugs, infection, structural; thiamine

Complications

- Hypoxia, aspiration, acidosis, hyperthermia, rhabdomyolysis, neuronal injury

Key point: Under-dosed benzodiazepines are the commonest error.

Ch 41 · Neuroleptic malignant syndrome

Features

- D2 blockade; hyperthermia, lead-pipe rigidity, altered mental state, autonomic lability; high CK

Differentials

- Serotonin syndrome (clonus, rapid onset), malignant hyperthermia (anaesthesia), anticholinergic

Management

- Stop drug; cooling, fluids, treat rhabdomyolysis
- Benzodiazepines, bromocriptine, dantrolene; paralyse if extreme; ECT if refractory

Key point: NMS evolves over days; serotonin syndrome over hours.

Ch 42 · Traumatic brain injury

Prevent secondary injury

- SBP ≥ 100–110, PaO2 > 10 kPa, PaCO2 4.5–5, head up 30°, Na 140–150, normothermia
- Seizure prophylaxis 7 days; VTE prophylaxis once stable

Tiered ICP therapy (ICP < 22, CPP 60–70)

- Tier 1: sedation, CSF drainage, mannitol/hypertonic saline
- Tier 2: mild hypocapnia, paralysis, MAP challenge
- Tier 3: barbiturates, decompressive craniectomy (RESCUEicp), hypothermia not early (Eurotherm)

Drugs

- Steroids harmful (CRASH); TXA within 3 h for mild–moderate TBI (CRASH-3)

Key point: One episode of hypotension doubles mortality in severe TBI.

Ch 43 · Hypoxic-ischaemic brain injury

Mechanism

- Energy failure, excitotoxicity, reperfusion injury; vulnerable hippocampus, cortex, cerebellum

Prognostication (≥ 72 h)

- Absent pupillary/corneal reflexes, absent N20, malignant EEG, NSE > 60, imaging injury
- Status myoclonus; ≥ 2 concordant predictors

Next steps

- Honest family discussion; withdrawal with palliative care
- Assess brainstem death; organ donation referral

Key point: Sedation and hypothermia must be excluded before predicting poor outcome.

Ch 44 · Neuromuscular weakness in ICU

Causes

- GBS, myasthenic crisis, botulism, organophosphates, electrolytes, ICU-acquired weakness, cord lesion

Intubation (20/30/40)

- VC < 20 mL/kg, MIP weaker than −30, MEP < 40; bulbar failure; avoid suxamethonium

GBS treatment

- IVIG 2 g/kg or plasma exchange (no steroids)
- Dysautonomia, VTE prophylaxis, neuropathic pain, early tracheostomy, rehabilitation

Key point: Measure VC serially – SpO2 and PaCO2 deteriorate late.

Ch 58 · Zika virus and Guillain–Barré syndrome

Zika

- Aedes-borne flavivirus; also sexual/vertical; rash, conjunctivitis, arthralgia; microcephaly

Diagnosis

- RT-PCR blood/urine; serology cross-reacts with dengue (PRNT)
- GBS: CSF albuminocytological dissociation, nerve conduction studies

Management

- Respiratory monitoring, IVIG or PLEX, dysautonomia care
- Public health: mosquito avoidance, pregnancy advice

Key point: Post-infectious GBS follows the infection by 1–3 weeks.